Computational-Basis Success Probability under Phase-Flip Noise

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Abstract

In this study, we investigated whether computational-basis success probability alone can correctly evaluate a phase-corrupted Bell-state circuit. The ideal state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ was subjected to a phase-flip channel on the first qubit, and the error probability was swept from $p = 0$ to $p = 1$ in steps of 0.01. The computational-basis and X-basis correlated success probabilities, fidelity to $|\Phi^+\rangle$, and the signed and absolute values of the coherence element $\rho_{00,11}$ were compared. The success probability remained equal to one for all 101 simulated values, while the fidelity decreased from one to zero and $\mathrm{Re}(\rho_{00,11})$ changed from 0.5 to -0.5. At $p = 0.5$, the coherence vanished even though the selected success metric still reported a perfect result. A pre-registered finite-shot simulation then showed that more Z-basis samples do not repair this structural blind spot, whereas X-basis sampling uncertainty shrinks with shot count. Therefore, computational-basis success probability alone cannot certify the relative phase of the prepared Bell state.

Introduction

Noise limits the reliability of near-term quantum circuits and makes the choice of evaluation metric important [2]. A metric may correctly measure one property of a state while remaining insensitive to another. For the Bell state $|\Phi^+\rangle$, measurements in the computational basis produce 00 and 11 with equal probabilities. It is therefore natural to define both outcomes as successful and use $P(00) + P(11)$ to evaluate the circuit.

However, the same computational-basis outcomes are obtained from $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ and from the incoherent mixture $\frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$. These states have different relative phases or coherence even though their populations are identical. State fidelity and density-matrix elements contain information that is not present in the selected success probability [1]. In this work, a phase-flip channel represented by Kraus operators was used to isolate this difference [3]. The purpose was to determine whether a circuit can be judged as successful after the intended Bell state has been changed.

Methodology

Bell-state preparation

The computational basis was ordered as $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. Beginning with $|00\rangle$, a Hadamard gate was applied to the first qubit:

$$(H \otimes I)|00\rangle = H|0\rangle \otimes I|0\rangle \tag{1}$$

$$= \frac{|00\rangle + |10\rangle}{\sqrt{2}}. \tag{2}$$

A CNOT gate with the first qubit as the control transforms $|10\rangle \to |11\rangle$, and the output is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 1\end{pmatrix}^T. \tag{3}$$

The density matrix is obtained from the outer product:

$$\rho_+ = |\Phi^+\rangle\langle\Phi^+| \tag{4}$$

$$= \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|) \tag{5}$$

$$= \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|) \tag{6}$$

$$= \frac{1}{2}\begin{pmatrix}1 & 0 & 0 & 1\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 1 & 0 & 0 & 1\end{pmatrix}. \tag{7}$$

Thus, the population elements are $\rho_{00,00} = \rho_{11,11} = 1/2$, while the off-diagonal coherence elements are $\rho_{00,11} = \rho_{11,00} = 1/2$.

Phase-flip channel

The phase flip was applied to the first qubit using

$$Z_1 = Z \otimes I = \mathrm{diag}(1, 1, -1, -1). \tag{8}$$

For an error probability p, the channel is

$$\mathcal{E}_p(\rho) = (1-p)\rho + pZ_1\rho Z_1, \tag{9}$$

with Kraus operators

$$E_0 = \sqrt{1-p}\, I_4, \qquad E_1 = \sqrt{p}\, Z_1. \tag{10}$$

The completeness relation was verified explicitly:

$$E_0^\dagger E_0 + E_1^\dagger E_1 = (1-p)I_4^\dagger I_4 + pZ_1^\dagger Z_1 \tag{11}$$

$$= (1-p)I_4 + pZ_1^2 \tag{12}$$

$$= (1-p)I_4 + pI_4 = I_4. \tag{13}$$

Therefore, the map is trace preserving. Its operator-sum representation also guarantees complete positivity [3].

Since

$$Z_1|\Phi^+\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\Phi^-\rangle, \tag{14}$$

the noisy state is

$$\rho(p) = (1-p)\,|\Phi^+\rangle\langle\Phi^+| + p\,|\Phi^-\rangle\langle\Phi^-|\,. \tag{15}$$

Expanding the two projectors gives

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}. \tag{16}$$

For additional verification, let the diagonal entries of Z_1 be $(z_{00}, z_{01}, z_{10}, z_{11}) = (1, 1, -1, -1)$. Because $(Z_1\rho Z_1)_{ij} = z_i z_j \rho_{ij}$, the relevant density-matrix indices are

$$\rho_{00,00}(p) = (1-p)\frac{1}{2} + p(1)(1)\frac{1}{2} = \frac{1}{2}, \tag{17}$$

$$\rho_{11,11}(p) = (1-p)\frac{1}{2} + p(-1)(-1)\frac{1}{2} = \frac{1}{2}, \tag{18}$$

$$\rho_{00,11}(p) = (1-p)\frac{1}{2} + p(1)(-1)\frac{1}{2} = \frac{1-2p}{2}, \tag{19}$$

$$\rho_{11,00}(p) = \frac{1-2p}{2}. \tag{20}$$

Evaluation metrics

Three types of information were compared. First, the computational-basis success probability reads only the diagonal populations:

$$S(p) = P(00) + P(11) = \rho_{00,00}(p) + \rho_{11,11}(p) = 1. \tag{21}$$

Second, phase-sensitive X-basis correlation is measured with

$$\Pi_X = |{+}{+}\rangle\langle{+}{+}| + |{-}{-}\rangle\langle{-}{-}|\,, \tag{22}$$

where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. In the computational basis, the resulting success probability is

$$S_X(p) = \mathrm{Tr}(\Pi_X \rho(p)) \tag{23}$$

$$= \frac{1}{2}\left(\rho_{00,00} + \rho_{11,11} + \rho_{00,11} + \rho_{11,00}\right) \tag{24}$$

$$= 1 - p. \tag{25}$$

The code evaluates this trace from the projector. Third, because the reference state is pure, its fidelity is

$$F(p) = \langle\Phi^+|\,\rho(p)\,|\Phi^+\rangle \tag{26}$$

$$= (1-p)|\,\langle\Phi^+|\Phi^+\rangle\,|^2 + p|\,\langle\Phi^+|\Phi^-\rangle\,|^2 \tag{27}$$

$$= 1 - p. \tag{28}$$

Fourth, the coherence was evaluated using both its signed real part and its magnitude:

$$\mathrm{Re}(\rho_{00,11}) = \frac{1-2p}{2}, \qquad |\rho_{00,11}| = \frac{|1-2p|}{2}. \tag{29}$$

The signed value distinguishes a positive from a negative relative phase, whereas the magnitude only indicates the amount of coherence.

Finite-shot measurements and uncertainty

For N independent measurements, let K_B be the correlated-success count in basis $B \in \{Z, X\}$ and let $\widehat{S}_B = K_B/N$. The Z-basis model has $K_Z = N$ for every p: its accepted outcomes are always 00 or 11. In contrast,

$$K_X \sim \mathrm{Binomial}(N, 1-p), \qquad \mathbb{E}[\widehat{S}_X] = 1 - p. \tag{30}$$

Thus, Z-basis samples have the same correlated-success distribution at every phase-flip probability and cannot identify p, whereas the X-basis estimator contains phase-error information. Each observed binomial proportion was accompanied by a two-sided 95% Wilson score confidence interval. The Wilson interval remains finite and bounded at zero or all observed successes; it is a frequentist confidence interval, not a posterior probability.

Experiment Setup

The calculation was implemented in Python using NumPy. The phase-flip probability was swept over

$$p \in \{0, 0.01, 0.02, \ldots, 0.99, 1\}, \tag{31}$$

giving 101 deterministic density-matrix evaluations. For each value of p, the program applied the Kraus map and recorded $S(p)$, $S_X(p)$, $F(p)$, $\mathrm{Re}(\rho_{00,11})$, and $|\rho_{00,11}|$. Matplotlib was used to generate the figures, and the complete numerical sweep was stored in CSV format. No random sampling or quantum software development kit was used for this exact density-matrix sweep.

The separate finite-shot experiment was preregistered before its samples were generated. It used $p \in \{0, 0.25, 0.50, 0.75, 1\}$, shot budgets $N \in \{100, 1000, 10000\}$, both bases, and 100 repetitions per condition. The root seed was 20261030, with a recorded deterministic child seed for every row, yielding 3,000 rows. These are simulated binomial measurements from the stated model, not quantum-hardware observations.

The implementation was checked with parameterized pytest tests. These tests verify the analytic density matrix and metric relations, Kraus completeness, unit trace, Hermiticity, positive semidefiniteness, invalid input rejection, X-basis projector evaluation, and deterministic CSV regeneration.

Results

Representative numerical values from the 101-point sweep are shown in Table 1. The simulated values agree with Eqs. (21)–(29) to the stored precision.

Table 1: The success probability remains one even when fidelity and signed coherence show that the target Bell state has changed.

p	$S_Z(p)$	$S_X(p)$	$F(p)$	$\mathrm{Re}(\rho_{00,11})$	$\lvert\rho_{00,11}\rvert$
0.00	1.00	1.00	1.00	0.50	0.50
0.25	1.00	0.75	0.75	0.25	0.25
0.50	1.00	0.50	0.50	0.00	0.00
0.75	1.00	0.25	0.25	-0.25	0.25
1.00	1.00	0.00	0.00	-0.50	0.50

Figure 1 directly compares the two correlated-success metrics. The computational-basis success remains equal to one over the entire interval, while the X-basis diagnostic decreases linearly as the phase-flip probability increases.

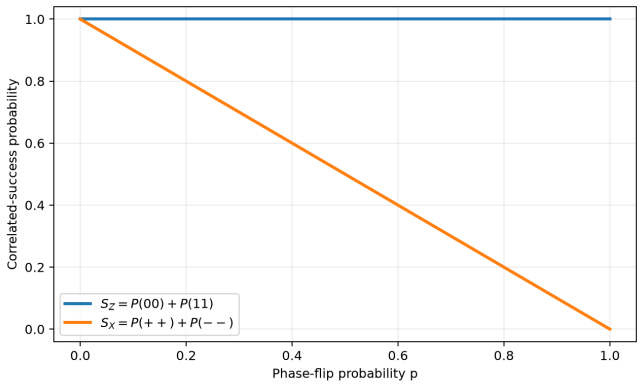


Figure 1: Computational-basis success is insensitive to phase flips, whereas the X-basis correlated-success diagnostic decreases from one to zero and exposes the degradation.

At $p = 0.5$, Eq. (16) becomes

$$\rho(0.5) = \frac{1}{2}\,|00\rangle\langle 00| + \frac{1}{2}\,|11\rangle\langle 11|\,. \tag{32}$$

The off-diagonal coherence has vanished, but $S(0.5) = 1$ whereas $S_X(0.5) = 0.5$. At $p = 1$, the state becomes $|\Phi^-\rangle\langle\Phi^-|$. It is fully coherent but orthogonal to the target, so $F(1) = 0$ while $S(1) = 1$.

Figure 2 shows why coherence magnitude alone is also insufficient. The magnitude decreases to zero at $p = 0.5$ and returns to 0.5 at $p = 1$, whereas the signed real part continues to -0.5 and exposes the phase reversal.

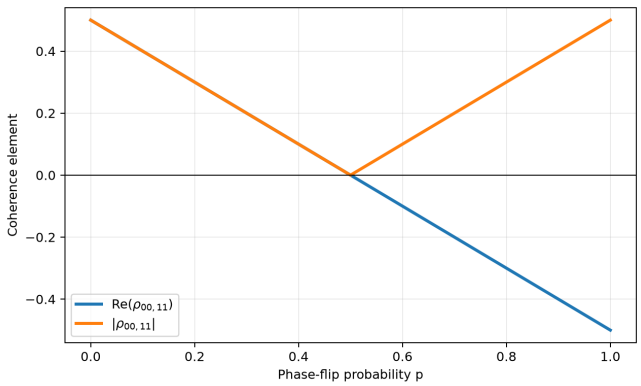


Figure 2: Signed coherence identifies the phase reversal at $p > 0.5$, while coherence magnitude alone cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$.

Figure 3 separates finite precision from metric blindness. At $p = 0.5$, every Z-basis sample has correlated-success estimate one regardless of shot budget, while the X-basis estimate is close to its analytic value of 0.5 and its Wilson interval narrows with additional shots. Across phase-flip probabilities, the X-basis empirical means track the analytic curve while the Z-basis empirical means remain one.

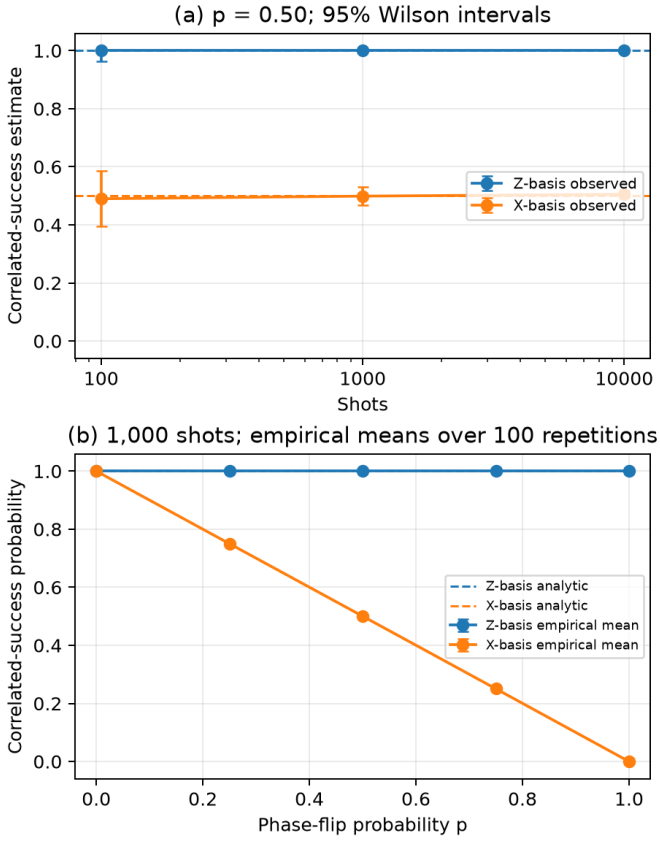


Figure 3: More shots do not repair the structural Z-basis blind spot: Z-basis correlated success remains one at $p = 0.5$ in every finite sample. Only X-basis sampling uncertainty shrinks with additional shots, and its empirical mean tracks the phase-sensitive analytic prediction.

Conclusion

This experiment demonstrated that computational-basis success probability alone can incorrectly judge a phase-corrupted Bell-state circuit as perfectly successful. The value of $P(00) + P(11)$ remained equal to one for every phase-flip probability, including the incoherent state at $p = 0.5$ and the orthogonal Bell state $|\Phi^-\rangle$ at $p = 1$. In contrast, fidelity and the signed coherence element detected the change in the quantum state. Therefore, a benchmark intended to verify Bell-state preparation must include phase-sensitive information such as the X-basis diagnostic rather than relying only on accepted computational-basis outcomes. Finite-shot sampling changes the precision of the X-basis estimate, but it cannot make the Z-basis metric sensitive to a relative phase that its accepted outcomes never test.

Discussion

The present work is an analytic density-matrix calculation reproduced by a deterministic classical simulation. It does not contain data from quantum hardware. The phase-flip channel is a simplified noise model and does not include state-preparation-and-measurement (SPAM) error, crosstalk, coherent calibration error, or drift. In addition, the noise channel was applied after an ideal Bell-state circuit and was not assigned to a specific pulse or time location inside a physical CNOT implementation. Consequently, the results do not estimate the CNOT error rate of IBM hardware or of any other real device.

The finite-shot extension adds binomial sampling uncertainty to the stated analytic probabilities, but it still omits hardware measurement mechanisms. The independent-shot assumption, fixed seed, and Wilson confidence intervals make the simulation reproducible; they do not turn the intervals into posterior probabilities or make them evidence about a physical device.

The compared metrics also have different limitations. Fidelity determines whether the state agrees with one specified target, but it is not a complete characterization of a general quantum process. Coherence magnitude detects the loss and recovery of coherence but removes its sign, so it cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$. A future experiment may compare phase-flip, bit-flip, and depolarizing channels, then evaluate the same metrics using a noise model tied to a specified circuit location. Hardware measurements would require separate calibration data and should not be interpreted using the present simplified model alone.

References

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[3] C. J. Wood, J. D. Biamonte, and D. G. Cory. Tensor networks and graphical calculus for open quantum systems. *Quantum Information and Computation*, 15(9–10):759–811, 2015.